

Comprehensive Exam-Oriented Study Guide

DATA AND VISUAL ANALYTICS (AL-603-B)

UNIT II

Advanced Data Analysis Techniques

B.Tech, VI Semester | RGPV Examination Scheme (GS)

DATA AND VISUAL ANALYTICS (AL-603-B)

Syllabus Topics Covered in This Unit

Statistical Hypothesis Generation and Testing

Chi-Square Test

t-Test

Analysis of Variance (ANOVA)

Correlation Analysis

Maximum Likelihood Test

Regression Modelling / Regression Analysis

Multivariate Analysis

Bayesian Modelling

Inference and Bayesian Network

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Unit Overview & Syllabus Map

Unit II builds directly on the statistical foundation from Unit I and takes it into the territory of actual hypothesis-driven decision making, model fitting, and multi-variable relationships. Where Unit I asked 'what does this data look like', Unit II asks 'can I trust a claim about this data, and can I build a model that explains or predicts it'. This unit is the analytical heart of the DVA syllabus and is very heavily tested — almost every RGPV paper carries at least 3-4 questions directly from this unit.

Official Syllabus Text

DEFINITION

Advance Data Analysis Techniques: Statistical hypothesis generation and testing, Chi-Square test, t-Test, Analysis of variance, Correlation analysis, Maximum likelihood test, Regression Modelling, Multivariate Analysis, Bayesian Modelling, Inference and Bayesian Network, Regression analysis.

EXAM TIP

Chi-Square, MLE, Regression, and Multivariate Analysis have appeared in BOTH available years — treat these four as guaranteed-to-repeat topics. ANOVA, Correlation, and Bayesian Networks have not appeared yet in these two papers but are explicitly in the syllabus, so still prepare them at a solid conceptual + one-numerical level as insurance.

LEARNING OBJECTIVES — By the end of this unit, you should be able to:

- ✓ State the general hypothesis-testing procedure and correctly identify Type I/Type II errors in a given scenario.
- ✓ Apply the Chi-Square test, t-test, and ANOVA to appropriate problems, including full worked numericals.
- ✓ Compute and interpret Pearson's correlation coefficient and distinguish correlation from regression.
- ✓ Derive a Maximum Likelihood Estimator for a simple distribution and explain its key properties.
- ✓ Build and interpret a simple/multiple linear regression model, including its assumptions and evaluation metrics.
- ✓ Explain multivariate analysis techniques (PCA, clustering, MANOVA) and Bayesian modelling/Bayesian networks at a conceptual level.

1. Statistical Hypothesis Generation and Testing

1.1 Introduction

Before any test (Chi-square, t-test, ANOVA) can be run, an analyst must first correctly generate a hypothesis — a precise, testable statement about a population parameter — and set up the testing framework. This topic is the conceptual umbrella under which every other test in this unit sits, so examiners frequently test its terminology as short-answer questions even when the main question is about a specific test.

1.2 Definitions

DEFINITION

Statistical Hypothesis: A precise, testable claim or assumption about a population parameter (such as a mean, proportion, or variance) that can be evaluated using sample data.

DEFINITION

Hypothesis Testing: A formal, structured procedure that uses sample evidence and probability theory to decide whether to reject or fail to reject a stated hypothesis about a population.

1.3 Hypothesis Generation — Where Hypotheses Come From

Hypothesis generation is the creative/analytical step of translating a business or research question into a statistically testable statement.

1. Start with a real-world question or observation (e.g., 'Do customers from Region A spend more than Region B?').
2. Translate it into a population parameter statement (e.g., mean spend of Region A = mean spend of Region B).
3. Formulate the Null Hypothesis (H_0) as the 'no difference / no effect / status quo' statement.
4. Formulate the Alternative Hypothesis (H_1 or H_a) as the statement you actually suspect/want evidence for.
5. Decide whether the alternative is one-tailed (a specific direction, e.g., 'greater than') or two-tailed (any difference).

1.4 Core Terminology (High-Yield for 2-3 Mark Questions)

Term	Meaning
Null Hypothesis (H_0)	The default assumption of no effect / no difference
Alternative Hypothesis (H_1)	The claim we look for evidence in favour of
Test Statistic	A single number (z , t , χ^2 , F) computed from sample data used to decide the test outcome
Significance level (α)	Probability of wrongly rejecting a true H_0 that we are willing to accept (commonly 0.05 or 0.01)
p-value	Probability of observing data as extreme as the sample, assuming H_0 is true
Critical value / Critical region	The threshold value(s) of the test statistic beyond which H_0 is rejected
Type I Error (α)	Rejecting a true H_0 — a 'false positive'

Term	Meaning
Type II Error (β)	Failing to reject a false H_0 — a 'false negative'
Power of a test ($1-\beta$)	Probability of correctly rejecting a false H_0
Degrees of Freedom (df)	Number of independent values free to vary in the final calculation of a statistic

1.5 Working Process — General Hypothesis Testing Procedure (Applies to Every Test in This Unit)

1. State H_0 and H_1 clearly.
2. Choose the significance level α (usually 0.05).
3. Select the appropriate test (z , t , χ^2 , F , etc.) based on data type, sample size, and what is being compared.
4. Compute the test statistic from the sample data.
5. Determine the critical value (from statistical tables) or compute the p-value.
6. Decision Rule: If $|\text{test statistic}| > \text{critical value}$ (or $p\text{-value} < \alpha$) $\rightarrow$ Reject H_0 . Otherwise $\rightarrow$ Fail to reject H_0 .
7. State the conclusion in the context of the original real-world question.

Decision Table (Type I / Type II Error)			
	H_0 is actually TRUE	H_0 is actually FALSE	
Reject H_0	Type I Error (α)	Correct decision (Power)	
Fail to Reject H_0	Correct decision	Type II Error (β)	

1.6 One-Tailed vs Two-Tailed Tests

Aspect	One-Tailed Test	Two-Tailed Test
H_1 statement	Parameter $>$ or $<$ a value (direction specified)	Parameter $\neq$ a value (no direction)
Critical region	Only one side of the distribution	Both sides of the distribution ($\alpha/2$ each)
Example	'New process yields MORE output than old'	'New process yields DIFFERENT output than old'

1.7 Real-World Examples

- A/B testing a website button color: H_0 = no difference in click-through rate; H_1 = there is a difference.
- Pharmaceutical trial: H_0 = new drug has no effect; H_1 = new drug reduces symptoms.
- Manufacturing QC: H_0 = mean bolt diameter = 5mm (spec); H_1 = mean diameter $\neq$ 5mm.

1.8 Advantages of a Structured Hypothesis Testing Framework

- Provides an objective, repeatable decision-making process instead of subjective judgment.
- Quantifies the risk of being wrong (via α and β) so decisions have a known error tolerance.
- Forms the statistical basis for A/B testing, quality control, and scientific research.

1.9 Limitations

- A statistically significant result is not always practically/business significant (large samples can make trivial differences 'significant').
- p-values are frequently misinterpreted as 'probability H0 is true' — this is incorrect; it is the probability of the data given H0.
- Choice of α is somewhat arbitrary and affects the conclusion.

1.10 Applications

- A/B testing in product analytics and marketing.
- Clinical trials and medical research.
- Quality control and Six Sigma manufacturing processes.
- Feature selection in machine learning (testing if a feature has a statistically significant relationship with the target).

MEMORY TIP

Remember 'HATCC' for the testing procedure: Hypotheses (H0/H1) → Alpha (significance level) → Test statistic → Critical value/p-value → Conclusion.

EXAM TIP

Whenever any question asks about Chi-square, t-test, or ANOVA, always begin your answer with these five generic hypothesis-testing steps before plugging in the test-specific formula — examiners award structural marks for this even before you reach the numeric answer.

2. Chi-Square Test

2.1 Introduction

The Chi-Square (χ^2) test is a non-parametric test — meaning it makes no assumption about the population following a normal distribution — used exclusively with categorical (nominal/ordinal) data. It is one of the most frequently asked topics in DVA exams, appearing in both available past papers.

2.2 Definition

DEFINITION

Chi-Square Test: A statistical hypothesis test that determines whether there is a statistically significant difference between observed frequencies and expected frequencies in one or more categories, used for testing goodness-of-fit or independence between categorical variables.

2.3 Two Major Types

- Chi-Square Goodness-of-Fit Test:
- Tests whether a single categorical variable's observed distribution matches an expected/theoretical distribution.
- Chi-Square Test of Independence:
- Tests whether two categorical variables are statistically independent of each other, using a contingency (cross-tab) table.

2.4 Formula

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

O_i = Observed frequency in category i

E_i = Expected frequency in category i

df (goodness-of-fit) = $k - 1$

df (independence, $r \times c$ table) = $(r - 1) \times (c - 1)$

2.5 Methodology — Step-by-Step

1. State H_0 (no association / observed fits expected) and H_1 (association exists / observed does not fit).
2. Build the contingency table of observed frequencies (for independence) or list expected proportions (for goodness-of-fit).
3. Compute expected frequency for each cell: $E_{ij} = (\text{Row Total} \times \text{Column Total}) / \text{Grand Total}$ (for independence test).
4. Compute $\chi^2 = \sum (O-E)^2/E$ across every cell/category.
5. Determine degrees of freedom and look up the critical χ^2 value at the chosen α (commonly 0.05).
6. Compare: if computed $\chi^2 > \text{critical } \chi^2$, reject H_0 (significant association exists); otherwise fail to reject H_0 .

2.6 Worked Example — Test of Independence

Question: Is product preference (Product A / Product B) independent of customer region (North / South)?
Observed data:

Region	Prefers A	Prefers B	Row Total
North	30	20	50
South	20	30	50
Column Total	50	50	100

Expected frequency for North/A = $(50 \times 50) / 100 = 25$. Similarly, all four expected cells work out to 25 (since row and column totals are equal here).

$$\chi^2 = (30-25)^2/25 + (20-25)^2/25 + (20-25)^2/25 + (30-25)^2/25 \\ = 25/25 + 25/25 + 25/25 + 25/25 = 1 + 1 + 1 + 1 = 4$$

df = $(2-1) \times (2-1) = 1$; Critical value at $\alpha=0.05$, df=1 is 3.841
Since computed χ^2 (4) > critical value (3.841) -> Reject H_0
Conclusion: Product preference IS associated with region.

2.7 Assumptions of the Chi-Square Test

- Data must be frequencies/counts, not percentages or raw measurements.
- Categories must be mutually exclusive.
- Expected frequency in each cell should generally be at least 5 for the test to be reliable.
- Observations must be independent of each other.

2.8 Advantages

- Works with categorical/nominal data where mean-based tests cannot be applied.
- Simple to compute and widely supported in statistical/BI software.
- Useful for both goodness-of-fit and association testing.

2.9 Limitations

- Sensitive to small expected frequencies (test becomes unreliable if $E_i < 5$ in many cells).
- Only tells you THAT an association exists, not the strength or direction of the relationship (Cramér's V is used for strength).
- Cannot be used directly on continuous data without first binning it into categories.

2.10 Applications

- Market research (association between demographic variables and preferences).
- Genetics (testing observed vs Mendelian expected ratios).
- Feature selection for categorical variables in machine learning pipelines.
- A/B testing of categorical outcomes (e.g., converted vs not converted, by variant).

MEMORY TIP

Chi-Square memory tip: 'OMG, Everyone!' — Observed Minus expected, squared, over Expected, then sum it all.

EXAM TIP

For Chi-square numericals, always show the contingency table, the expected-frequency formula application for at least one cell, the full χ^2 computation, the df calculation, and an explicit comparison-based conclusion — RGPV awards marks at each of these five checkpoints.

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3. t-Test

3.1 Introduction

The t-test is a parametric test used to compare means when the sample size is small (typically $n < 30$) and/or the population standard deviation is unknown. It relies on the t-distribution, which has heavier tails than the normal distribution to account for the extra uncertainty of estimating the standard deviation from a small sample. This topic was asked as a full numerical (potato yield example) in May 2024.

3.2 Definition

DEFINITION

t-Test: A statistical hypothesis test that uses the Student's t-distribution to determine whether there is a significant difference between the mean of a sample and a known value, or between the means of two groups, when the population standard deviation is unknown.

3.3 Types of t-Tests

Type	Purpose	Formula (test statistic)
One-sample t-test	Compare sample mean to a known/hypothesized population mean μ	$t = (\bar{x} - \mu) / (s/\sqrt{n})$
Independent (two-sample) t-test	Compare means of two independent, unrelated groups	$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$
Paired t-test	Compare means of the same subjects measured twice (before/after)	$t = \bar{d} / (sd/\sqrt{n})$, where d = paired differences

3.4 Methodology — Step-by-Step (One-Sample t-Test)

1. State $H_0: \mu = \mu_0$ (population mean equals a claimed value) and $H_1: \mu \neq \mu_0$ (or $>$ / $<$ for one-tailed).
2. Compute the sample mean $\bar{x}$ and sample standard deviation s from the data.
3. Compute degrees of freedom: $df = n - 1$.
4. Compute the t-statistic: $t = (\bar{x} - \mu_0) / (s/\sqrt{n})$.
5. Find the critical t-value from the t-table at the chosen α and df (or compute the p-value).
6. If $|t_{\text{computed}}| > t_{\text{critical}} \rightarrow$ reject H_0 ; otherwise fail to reject H_0 .

3.5 Worked Example (RGPV-Style Numerical, as asked May 2024)

Potato yield from 12 farms: 21.5, 24.5, 18.5, 17.2, 14.5, 23.2, 22.1, 20.5, 19.4, 18.1, 24.1, 18.5. Standard yield $\mu = 20$. Test if the yield from these farms is significantly different (better) than the standard, at $\alpha = 0.05$.

1. $H_0: \mu = 20$ (farm yield equals standard). $H_1: \mu > 20$ (farm yield is significantly better) — one-tailed test.
2. Sample mean $\bar{x} = \Sigma x / n = 241.6 / 12 = 20.13$ (approx).
3. Compute sample standard deviation s using $s^2 = \Sigma(x - \bar{x})^2 / (n-1)$; this works out to approximately $s \approx 2.99$.
4. $t = (\bar{x} - \mu) / (s/\sqrt{n}) = (20.13 - 20) / (2.99/\sqrt{12}) \approx 0.13 / 0.863 \approx 0.15$.

5. $df = 12 - 1 = 11$. Critical t-value at $\alpha = 0.05$ (one-tailed), $df = 11$ is approximately 1.796.
6. Since computed $t (\approx 0.15) < \text{critical } t (1.796) \rightarrow \text{Fail to reject } H_0$.
7. Conclusion: The potato yield from these farms is NOT significantly better than the standard yield.

EXAM TIP

In numerical t-test answers, even if your arithmetic is slightly off, clearly showing all six steps above (hypotheses $\rightarrow$ mean $\rightarrow$ SD $\rightarrow$ t-statistic $\rightarrow$ critical value $\rightarrow$ conclusion) secures the majority of the marks.

3.6 Assumptions of the t-Test

- Data is approximately normally distributed (more important for small n).
- Observations are independent of each other.
- For the two-sample t-test, variances of the two groups are assumed roughly equal (unless using Welch's correction).

3.7 t-Test vs Z-Test (Common Exam Comparison)

Aspect	t-Test	Z-Test
Sample size	Small ($n < 30$ typically)	Large ($n \geq 30$)
Population SD (σ)	Unknown, estimated from sample (s)	Known
Distribution used	t-distribution (heavier tails)	Standard normal distribution

3.8 Advantages

- Works well even with small sample sizes.
- Does not require knowledge of the population standard deviation.
- Widely available and simple to compute in Excel/Python/R.

3.9 Limitations

- Assumes approximate normality — unreliable for heavily skewed small samples.
- Only compares means of one or two groups (for 3+ groups, ANOVA is required instead).
- Sensitive to outliers, which can distort both the mean and standard deviation.

3.10 Applications

- Comparing before/after performance (paired t-test) in process improvement studies.
- Comparing two marketing campaigns' average conversion value (independent t-test).
- Quality control: testing if a batch's mean measurement matches a specification (one-sample t-test).

MEMORY TIP

t-Test memory tip: 't' stands for 'tiny samples' — use it when your sample is small and you don't know the population's true spread.

4. Analysis of Variance (ANOVA)

4.1 Introduction

While the t-test can only compare the means of two groups at a time, real analytical problems often involve comparing three or more groups (e.g., sales across four regions, test scores across three teaching methods). Running multiple t-tests pairwise inflates the overall Type I error rate. ANOVA solves this by testing all group means simultaneously using a single test based on variance comparison.

4.2 Definition

DEFINITION

Analysis of Variance (ANOVA): A statistical technique that tests whether the means of three or more groups are significantly different from each other by comparing the variance BETWEEN groups to the variance WITHIN groups, using the F-distribution.

4.3 The Core Idea

ANOVA's key insight is that if group means are truly different, the variability between the group means will be large relative to the natural variability within each group. This ratio is captured by the F-statistic.

$$F = \frac{\text{Variance BETWEEN groups (MSB)}}{\text{Variance WITHIN groups (MSW)}}$$

Large F → between-group differences dominate random within-group noise → means likely differ

F near 1 → between-group spread is no more than ordinary random noise → means likely equal

4.4 Types of ANOVA

Type	Use Case
One-Way ANOVA	Compares means across groups defined by ONE categorical factor (e.g., 3 teaching methods)
Two-Way ANOVA	Compares means across groups defined by TWO categorical factors simultaneously, and can test their interaction (e.g., teaching method AND gender)

4.5 One-Way ANOVA — Hypotheses

- $H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$ (all group means are equal)
- H_1 : At least one group mean is different from the others

4.6 Methodology — Step-by-Step

1. State H_0 and H_1 as above.
2. Compute the grand mean (mean of all observations across all groups).
3. Compute Sum of Squares Between groups (SSB) — how far each group's mean is from the grand mean, weighted by group size.

4. Compute Sum of Squares Within groups (SSW) — how far each observation is from its own group's mean.
5. Compute Mean Squares: $MSB = SSB / (k-1)$ and $MSW = SSW / (N-k)$, where k = number of groups, N = total observations.
6. Compute $F = MSB / MSW$.
7. Compare computed F with the critical F -value from the F -table at $df_1=(k-1)$, $df_2=(N-k)$ and chosen α .
8. If F computed $>$ F critical $\rightarrow$ reject H_0 (at least one group mean differs); if significant, follow up with a post-hoc test (e.g., Tukey's HSD) to find WHICH groups differ.

4.7 Standard ANOVA Summary Table (Exam-Ready Format)

Source of Variation	Sum of Squares (SS)	df	Mean Square (MS)	F
Between Groups	SSB	$k - 1$	$MSB = SSB/(k-1)$	$F = MSB/MSW$
Within Groups	SSW	$N - k$	$MSW = SSW/(N-k)$	—
Total	$SST = SSB + SSW$	$N - 1$	—	—

EXAM TIP

Always reproduce this exact ANOVA summary table in your answer — it is the single most recognizable 'expected format' examiners look for, even if you cannot complete every numeric cell.

4.8 Assumptions of ANOVA

- Observations are independent.
- Each group's data is approximately normally distributed.
- Homogeneity of variance — all groups have approximately equal variance (tested using Levene's test).

4.9 Advantages

- Compares 3+ groups in a single test, controlling the overall Type I error rate.
- More statistically efficient than running many pairwise t-tests.
- Two-way ANOVA can additionally reveal interaction effects between two factors.

4.10 Limitations

- Only tells you that SOME group differs, not which one — requires a post-hoc test for pairwise detail.
- Sensitive to violations of the equal-variance assumption.
- Less powerful/valid when sample sizes across groups are very unequal or heavily skewed.

4.11 Applications

- Comparing crop yield across multiple fertilizer treatments (agricultural research).
- Comparing average sales performance across multiple store regions.
- Comparing effectiveness of multiple drug dosages in clinical trials.

MEMORY TIP

ANOVA memory tip: 'Analysis Not Of Variances Alone' — ANOVA compares MEANS, even though it does the comparison BY analyzing variance.

5. Correlation Analysis

5.1 Introduction

Correlation analysis measures the strength and direction of the linear relationship between two quantitative variables. It is often the first step before building a regression model, since a strong correlation suggests a predictive relationship may exist (though correlation alone never proves causation).

5.2 Definition

DEFINITION

Correlation: A statistical measure that describes the extent to which two variables change together — i.e., the strength and direction of their linear association — typically quantified by the correlation coefficient r , which ranges from -1 to $+1$.

5.3 Pearson's Correlation Coefficient (Most Important Formula)

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{[\sum (x_i - \bar{x})^2] \times [\sum (y_i - \bar{y})^2]}}$$

$r = +1$ → perfect positive linear relationship

$r = 0$ → no linear relationship

$r = -1$ → perfect negative linear relationship

5.4 Interpreting the Strength of r

Range of $ r $	Strength of Relationship
0.00 – 0.19	Very weak / negligible
0.20 – 0.39	Weak
0.40 – 0.59	Moderate
0.60 – 0.79	Strong
0.80 – 1.00	Very strong

5.5 Types of Correlation

- Positive correlation:
 - Both variables increase together (e.g., height and weight).
- Negative correlation:
 - One variable increases as the other decreases (e.g., price and demand).
- Zero/No correlation:
 - No consistent linear pattern between the variables.
- Simple correlation:
 - Relationship between exactly two variables.
- Multiple correlation:
 - Relationship between one variable and a group of two or more other variables together.

- Partial correlation:
- Relationship between two variables while controlling for the effect of a third variable.

5.6 Correlation vs Regression (Frequently Confused — Exam Favourite)

Aspect	Correlation	Regression
Purpose	Measures strength/direction of a relationship	Models/predicts one variable from another
Symmetry	Symmetric — $r(x,y) = r(y,x)$	Asymmetric — predicting y from x differs from predicting x from y
Output	A single number r (–1 to +1)	An equation (e.g., $y = a + bx$) usable for prediction
Causation	Does not imply causation	Also does not imply causation, but models a directional dependency

5.7 Methodology — Step-by-Step (Pearson's r)

1. Collect paired observations (x_i, y_i) for the two variables.
2. Compute the mean of x ($\bar{x}$) and mean of y ($\bar{y}$).
3. Compute the deviations ($x_i - \bar{x}$) and ($y_i - \bar{y}$) for each pair.
4. Compute the numerator: sum of the products of the deviations.
5. Compute the denominator: square root of (sum of squared x-deviations $\times$ sum of squared y-deviations).
6. Divide numerator by denominator to get r .
7. Interpret sign (direction) and magnitude (strength) using the table above.

5.8 Spearman's Rank Correlation (For Ordinal Data)

DEFINITION

Spearman's Rank Correlation (ρ): A non-parametric measure of correlation that uses the RANKS of the data rather than raw values, suitable for ordinal data or when the relationship is monotonic but not strictly linear.

Formula: $\rho = 1 - [6 \sum d_i^2 / (n(n^2-1))]$

where d_i = difference between the ranks of each paired observation, n = number of pairs

5.9 Advantages

- Simple, single-number summary of relationship strength and direction.
- Quick first step in exploratory data analysis before modelling.
- Basis for feature selection in predictive modelling (drop highly correlated redundant features).

5.10 Limitations

- Pearson's r only captures LINEAR relationships — a strong curved relationship can show $r \approx 0$.
- Highly sensitive to outliers.
- Correlation does not imply causation — a classic exam point (e.g., ice cream sales correlate with drowning deaths, both driven by hot weather).

5.11 Applications

- Feature selection and multicollinearity detection in machine learning.
- Finance (correlation between stock returns for portfolio diversification).
- Market research (relationship between advertising spend and sales).

MEMORY TIP

Correlation memory tip: 'Correlation Cannot Cause' — always mention that correlation never proves causation; this single line frequently earns bonus credit in DVA answers.

6. Maximum Likelihood Test (Maximum Likelihood Estimation)

6.1 Introduction

Maximum Likelihood Estimation (MLE) is one of the most fundamental parameter-estimation techniques in statistics and machine learning, and has been directly asked in both available RGPV papers. Rather than testing a hypothesis about a known parameter, MLE answers a different question: 'given this observed data, what parameter values make this data most probable?'

6.2 Definition

DEFINITION

Maximum Likelihood Estimation (MLE): A method for estimating the parameters of an assumed probability distribution by finding the parameter values that maximize the likelihood function — the probability of observing the given sample data as a function of the unknown parameters.

6.3 The Likelihood Function

If we assume the data $x_1, x_2, \dots, x_n$ are independent and identically distributed draws from a distribution with probability density/mass function $f(x; \theta)$, the likelihood function is:

$$L(\theta) = f(x_1; \theta) \times f(x_2; \theta) \times \dots \times f(x_n; \theta) = \prod f(x_i; \theta)$$

Taking the log (log-likelihood) converts the product into an easier-to-differentiate sum:

$$\ell(\theta) = \log L(\theta) = \sum \log f(x_i; \theta)$$

6.4 Methodology — Step-by-Step

1. Assume a probability distribution/model for the data with unknown parameter(s) θ (e.g., Normal, Bernoulli, Poisson).
2. Write the likelihood function $L(\theta)$ as the joint probability of observing the sample, treating θ as the unknown variable.
3. Take the natural log to get the log-likelihood $\ell(\theta)$, which turns the product into a sum (much easier to differentiate).
4. Differentiate $\ell(\theta)$ with respect to θ and set the derivative equal to zero: $d\ell/d\theta = 0$.
5. Solve the resulting equation for θ to obtain the maximum likelihood estimate $\hat{\theta}$.
6. Verify $\hat{\theta}$ is indeed a maximum using the second derivative test ($d^2\ell/d\theta^2 < 0$).

6.5 Worked Example — MLE for a Normal Distribution

For n independent observations assumed to be $\text{Normal}(\mu, \sigma^2)$, maximizing the log-likelihood with respect to μ and σ^2 gives:

MLE of mean:	$\hat{\mu} = (1/n) \sum x_i = \bar{x}$	(the ordinary sample mean)
MLE of variance:	$\hat{\sigma}^2 = (1/n) \sum (x_i - \bar{x})^2$	(note: divides by n , NOT $n-1$)

This is a classic exam point: the MLE variance estimator is BIASED (slightly underestimates true population variance) precisely because it divides by n instead of $n-1$; the sample variance formula from Unit I (with $n-1$, 'Bessel's correction') is the bias-corrected version.

6.6 Worked Example — MLE for a Bernoulli/Binomial Proportion

If we observe n coin flips with k heads, and each flip is $\text{Bernoulli}(p)$, the MLE of p is simply the observed proportion:

$$\hat{p} = k / n \quad (\text{e.g., } 7 \text{ heads out of } 10 \text{ flips} \rightarrow \hat{p} = 0.7)$$

6.7 Properties of Maximum Likelihood Estimators

- Consistency:
- As sample size $n \rightarrow \infty$, the MLE converges to the true parameter value.
- Asymptotic Normality:
- For large n , the sampling distribution of the MLE is approximately normal.
- Efficiency:
- Among unbiased estimators, MLE achieves the lowest possible variance for large samples (achieves the Cramér–Rao lower bound).
- Invariance:
- If $\hat{\theta}$ is the MLE of θ , then $g(\hat{\theta})$ is the MLE of $g(\theta)$ for any function g .

6.8 Advantages

- Provides a systematic, mathematically principled way to fit model parameters to data.
- Asymptotically efficient — becomes the best possible unbiased estimator as sample size grows.
- Directly generalizes to complex models (logistic regression, hidden Markov models, neural network training via cross-entropy loss, which is derived from MLE).

6.9 Limitations

- Can be biased for small sample sizes (as seen with the variance example above).
- Requires assuming a specific probability distribution — a poor distributional assumption leads to poor estimates.
- The likelihood equation may not always have a closed-form solution, requiring numerical optimization (e.g., gradient descent, Newton-Raphson).
- Sensitive to outliers if the assumed distribution is not robust (e.g., Normal MLE is sensitive to extreme values).

6.10 Applications

- Fitting probability distributions to observed data (curve fitting).
- Logistic regression coefficient estimation.
- Training many machine learning models (linear regression under Gaussian noise assumptions, Naive Bayes classifiers, Hidden Markov Models).
- Actuarial science and risk modelling.

MEMORY TIP

MLE memory tip: 'Pick the parameter that makes YOUR data look the most likely, not the most typical' — MLE always asks 'what θ best explains what I actually saw?'

EXAM TIP

For MLE questions, always show: (1) the assumed distribution and its pdf, (2) the likelihood function as a product, (3) the log-likelihood as a sum, (4) the derivative set to zero, (5) the final closed-form estimator. This 5-step structure matches exactly what was asked in both 2023 and 2024 papers.

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7. Regression Modelling and Regression Analysis

7.1 Introduction

Regression is arguably the single most important predictive modelling technique in data analytics — it models the relationship between a dependent (outcome) variable and one or more independent (predictor) variables, enabling both explanation and prediction. This topic has appeared in both papers, once focused on variable types in regression modelling and once on simple/multiple regression directly.

7.2 Definitions

DEFINITION

Regression Analysis: A set of statistical techniques for estimating the relationships between a dependent variable and one or more independent variables, typically used to understand how the dependent variable changes when any independent variable is varied, and to predict future values.

DEFINITION

Regression Modelling: The broader process of selecting, fitting, validating, and refining a regression model — including choosing variables, checking assumptions, and evaluating model fit — not just computing the coefficients.

7.3 Types of Variables in Regression (Asked Directly in May 2023)

Variable Role	Also Called	Description	Example
Dependent Variable	Response / Outcome / Target (y)	The variable being predicted or explained	House price
Independent Variable(s)	Predictor / Explanatory / Feature (x)	The variable(s) used to predict/explain y	House size, location, age
Confounding Variable	Lurking variable	An unmeasured variable influencing both x and y, creating a misleading association	City wealth affecting both house price and school quality
Dummy/Indicator Variable	Binary/categorical encoding	A 0/1 variable representing a categorical predictor's level	IsCorner_House = 1 or 0

7.4 Simple Linear Regression

Models the relationship between exactly one independent variable and one dependent variable as a straight line.

$$y = a + bx + \epsilon$$

y = dependent variable (predicted)

x = independent variable

a = intercept (value of y when x = 0)

b = slope (change in y per unit change in x)

ϵ = random error term

The coefficients a and b are typically estimated using the Ordinary Least Squares (OLS) method, which minimizes the sum of squared residuals (the vertical distances between actual and predicted y values).

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$a = \bar{y} - b \bar{x}$$

7.5 Multiple Linear Regression

Extends simple regression to model the dependent variable using two or more independent variables simultaneously, capturing more complex real-world relationships.

$$y = a + b_1 x_1 + b_2 x_2 + \dots + b_k x_k + \epsilon$$

Each b_i represents the change in y per unit change in x_i , HOLDING ALL OTHER predictors constant.

7.6 Methodology — Building a Regression Model (Working Process)

1. Define the dependent variable and candidate independent variables based on the business/research question.
2. Explore the data — check correlations, scatter plots, and screen for multicollinearity among predictors.
3. Split the data into training and testing sets (in a predictive-modelling context).
4. Fit the model using Ordinary Least Squares (or Maximum Likelihood, which gives identical results under Gaussian error assumptions).
5. Evaluate model fit using R^2 , Adjusted R^2 , and residual analysis.
6. Check regression assumptions (linearity, independence, homoscedasticity, normal residuals).
7. Interpret coefficients and use the model for prediction or explanation.

7.7 Assumptions of Linear Regression (Frequently Tested)

- Linearity: the relationship between predictors and the outcome is linear.
- Independence of errors: residuals are not correlated with each other.
- Homoscedasticity: residuals have constant variance across all levels of the predictors.
- Normality of residuals: errors are approximately normally distributed.
- No (or low) multicollinearity: independent variables are not highly correlated with each other.

7.8 Evaluating Model Fit

Metric	Meaning
R^2 (Coefficient of Determination)	Proportion of variance in y explained by the model (0 to 1; higher is better fit)
Adjusted R^2	R^2 adjusted for the number of predictors — penalizes adding useless variables
RMSE (Root Mean Squared Error)	Average magnitude of prediction error, in the same units as y
p-value of coefficients	Tests whether each predictor's relationship with y is statistically significant

7.9 Types of Regression Models (Beyond Linear — Good for Bonus Marks)

- Logistic Regression:
- Used when the dependent variable is categorical/binary (e.g., churn: yes/no); models the log-odds of the outcome.
- Polynomial Regression:
- Fits a curved (non-linear) relationship by including powers of x (x^2 , x^3 , ...).
- Ridge/Lasso Regression:
- Regularized regression variants that penalize large coefficients to prevent overfitting, useful with many correlated predictors.

7.10 Advantages

- Highly interpretable — coefficients directly show the effect size and direction of each predictor.
- Computationally efficient and well-understood, with strong theoretical guarantees.
- Forms the mathematical basis for many more advanced models (GLMs, neural network output layers).

7.11 Limitations

- Only captures linear relationships unless explicitly extended (polynomial terms, transformations).
- Sensitive to outliers, which can disproportionately pull the fitted line.
- Multicollinearity among predictors makes individual coefficient estimates unstable and hard to interpret.
- Assumes the specified functional form is correct — a poorly specified model gives misleading results even with a high R^2 .

7.12 Applications

- Sales forecasting based on advertising spend, seasonality, and pricing.
- Real estate price prediction based on size, location, and amenities.
- Risk scoring in finance and insurance (e.g., credit scoring with logistic regression).
- Trend analysis and 'what-if' scenario planning in business intelligence dashboards.

MEMORY TIP

Regression memory tip: 'LINE' for assumptions — Linearity, Independence, Normality of residuals, Equal variance (homoscedasticity).

EXAM TIP

For 'types of variables in regression' (May 2023 style question), always distinguish Dependent vs Independent first, then mention confounding and dummy variables as the 'extra depth' examiners reward beyond the basic answer.

8. Multivariate Analysis

8.1 Introduction

Real datasets rarely involve just one or two variables — multivariate analysis is the family of statistical techniques designed to analyze data involving three or more variables simultaneously, capturing complex interactions that univariate or bivariate analysis would miss. This is one of the most consistently tested Unit 2 topics, appearing in both available papers.

8.2 Definition

DEFINITION

Multivariate Analysis: A set of statistical techniques used to analyze data that involves more than two variables simultaneously, with the goal of understanding relationships, patterns, and structure among multiple variables at once.

8.3 Univariate vs Bivariate vs Multivariate (Foundational Comparison)

Type	Number of Variables	Typical Technique	Example
Univariate	1	Mean, median, histogram, box plot	Distribution of employee salaries
Bivariate	2	Correlation, scatter plot, simple regression	Relationship between salary and experience
Multivariate	3+	Multiple regression, PCA, cluster analysis, MANOVA	Salary explained by experience, education, AND department

8.4 Major Multivariate Techniques

- Multiple Regression:
- Predicting one dependent variable using several independent variables (covered in Section 7).
- Principal Component Analysis (PCA):
- Reduces a large number of correlated variables into a smaller set of uncorrelated 'principal components' that capture most of the original variance.
- Factor Analysis:
- Identifies underlying latent factors that explain the correlations among a set of observed variables.
- Cluster Analysis:
- Groups observations into clusters such that items within a cluster are more similar to each other than to items in other clusters (e.g., K-means).
- MANOVA (Multivariate ANOVA):
- Extends ANOVA to test differences across groups on TWO OR MORE dependent variables simultaneously.
- Discriminant Analysis:
- Predicts a categorical outcome (class label) using multiple continuous predictor variables.
- Canonical Correlation Analysis:
- Examines the relationship between two SETS of variables (rather than two single variables).

8.5 Principal Component Analysis (PCA) — Deep Dive (High-Value Topic)

PCA is the most commonly examined multivariate technique because it elegantly solves the 'curse of dimensionality' — too many correlated variables making analysis and visualization difficult.

1. Standardize the data (mean 0, variance 1) so variables with larger scales don't dominate.
2. Compute the covariance (or correlation) matrix of the standardized variables.
3. Compute the eigenvalues and eigenvectors of this covariance matrix.
4. Sort eigenvectors by their corresponding eigenvalues in descending order — each eigenvector is a 'principal component' direction, and its eigenvalue represents how much variance it explains.
5. Select the top k principal components that together explain a satisfactory proportion of total variance (e.g., 90%).
6. Project the original data onto these k components to get a reduced-dimension dataset.

Original: 10 correlated variables (X1...X10)
|
v PCA
Reduced: 2-3 uncorrelated Principal Components (PC1, PC2, PC3)
that capture ~85-95% of the original variance

8.6 Working Process — General Multivariate Analysis Workflow

1. Define the research question and identify all relevant variables.
2. Clean and preprocess the data (handle missing values, standardize scales).
3. Explore relationships using a correlation matrix and pairwise scatter plots.
4. Select the appropriate multivariate technique based on the goal (prediction → regression; grouping → clustering; dimension reduction → PCA/factor analysis; group comparison → MANOVA).
5. Fit the model and validate assumptions specific to the chosen technique.
6. Interpret results in the context of the original business/research question.

8.7 Advantages of Multivariate Analysis

- Captures complex, realistic relationships that involve many factors acting together.
- Reduces dimensionality, making large datasets easier to visualize and interpret (via PCA/factor analysis).
- Controls for confounding variables by including them explicitly in the model.

8.8 Limitations

- Requires larger sample sizes to reliably estimate more parameters.
- Results (especially PCA components, factors) can be harder to interpret than simple univariate/bivariate statistics.
- Computationally more intensive, and more sensitive to violated assumptions (e.g., multicollinearity, non-normality).

8.9 Applications

- Customer segmentation in marketing (cluster analysis on purchase behaviour variables).
- Image compression and facial recognition (PCA-based dimensionality reduction, e.g., eigenfaces).
- Genomics — analyzing thousands of gene expression variables together.

- Finance — portfolio risk modelling using multiple correlated asset returns.

MEMORY TIP

Multivariate memory tip: 'More variables, More patterns, More power — but More complexity too' — always mention this trade-off in exam answers for balanced marks.

EXAM TIP

For 'Multivariate Exploration of Data' questions, structure your answer as: define multivariate analysis → compare with univariate/bivariate in a table → list 3-4 named techniques (PCA, cluster analysis, MANOVA, multiple regression) with one line each → mention one real-world application.

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9. Bayesian Modelling

9.1 Introduction

Bayesian modelling represents a fundamentally different philosophy of statistics compared to the 'frequentist' approach used in Chi-square, t-tests, and classical regression. Instead of treating a population parameter as a fixed unknown constant, the Bayesian approach treats it as a random variable with its own probability distribution that gets updated as new evidence arrives. This topic appeared directly in the May 2024 paper.

9.2 Definition

DEFINITION

Bayesian Modelling: A statistical approach that combines prior knowledge or belief about a parameter (the Prior) with observed data (via the Likelihood) to produce an updated belief about the parameter (the Posterior), using Bayes' Theorem.

9.3 Bayes' Theorem — The Core Formula

$$P(\theta \mid \text{Data}) = \frac{P(\text{Data} \mid \theta) \times P(\theta)}{P(\text{Data})}$$

$P(\theta)$	= Prior	– belief about θ before seeing the data
$P(\text{Data} \mid \theta)$	= Likelihood	– probability of the data given θ (same as in MLE)
$P(\theta \mid \text{Data})$	= Posterior	– updated belief about θ after seeing the data
$P(\text{Data})$	= Evidence / Marginal likelihood	– normalizing constant

9.4 How Bayesian Modelling Works — Step-by-Step

1. Choose a Prior distribution $P(\theta)$ that reflects existing knowledge or assumptions about the parameter before observing any data (can be informative or 'flat'/uninformative if no prior knowledge exists).
2. Collect data and define the Likelihood function $P(\text{Data} \mid \theta)$ — the same likelihood used in MLE.
3. Apply Bayes' Theorem to combine Prior and Likelihood into the Posterior distribution $P(\theta \mid \text{Data})$.
4. Use the Posterior for inference: e.g., report the posterior mean/mode as a point estimate, or a credible interval as a range estimate.
5. As more data arrives, the current Posterior becomes the new Prior, and the process repeats — allowing continuous, incremental learning.

9.5 Worked Conceptual Example

Suppose a factory historically has a 2% defect rate (this is the Prior belief, $P(\text{defective}) = 0.02$). A new batch of 50 items is inspected and 5 are found defective (this is the observed Likelihood evidence). Bayesian updating combines the historical 2% prior with the new evidence from this batch to produce an updated (posterior) estimate of the true defect rate — which will sit somewhere between the old 2% belief and the new batch's raw 10% observed rate, weighted by how much data/confidence backs each source.

9.6 Bayesian vs Frequentist Approach (Classic Exam Comparison Table)

Aspect	Frequentist Approach	Bayesian Approach
View of parameter θ	Fixed, unknown constant	Random variable with its own distribution
Uses prior knowledge?	No	Yes — explicitly via the Prior
Output	Point estimate + p-value / confidence interval	Full posterior distribution + credible interval
Interpretation of interval	95% of such intervals would contain θ over repeated sampling	95% probability θ lies within this specific interval
Updating with new data	Requires re-running the analysis from scratch	Naturally incremental — posterior becomes new prior

9.7 Advantages of Bayesian Modelling

- Formally incorporates prior expert knowledge or historical data into the analysis.
- Provides an intuitive probability distribution over the parameter, not just a single point estimate.
- Naturally handles sequential/streaming data — updates smoothly as new evidence arrives.
- Works well even with small sample sizes, when informed by a reasonable prior.

9.8 Disadvantages / Limitations

- The choice of prior is subjective and can meaningfully influence results, especially with small samples.
- Computationally intensive for complex models (often requires simulation methods like Markov Chain Monte Carlo, MCMC).
- Harder to communicate to non-technical stakeholders compared to a simple p-value.

9.9 Applications

- Spam email filtering (Naive Bayes classifiers).
- Medical diagnosis — updating disease probability as test results arrive.
- Recommendation systems and online A/B testing (Bayesian bandits).
- Risk assessment in finance and insurance, combining historical data with expert judgment.

MEMORY TIP

Bayes memory tip: 'Prior x Likelihood = Posterior, divided by Evidence' — remember it as PLPE, and that the Posterior always becomes tomorrow's Prior.

10. Inference and Bayesian Network

10.1 Introduction

A Bayesian Network extends the single-variable idea of Bayesian modelling into a full graphical model that represents probabilistic relationships among MANY variables at once, making it possible to reason about complex systems with uncertain, interdependent factors — and to perform 'inference': answering probability queries using the network.

10.2 Definition

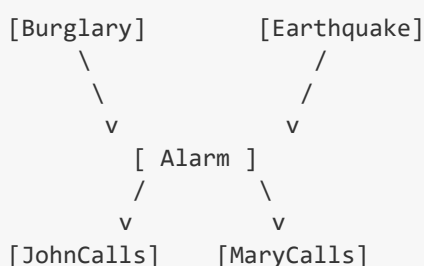
DEFINITION

Bayesian Network: A probabilistic graphical model represented as a Directed Acyclic Graph (DAG), where nodes represent random variables and directed edges represent conditional (causal or influential) dependencies between them, with each node holding a Conditional Probability Table (CPT) quantifying its dependence on its parent nodes.

10.3 Structure of a Bayesian Network

- Nodes:
- Represent random variables (can be discrete or continuous).
- Directed Edges:
- Represent direct probabilistic dependency; an edge from A to B means A influences B.
- Conditional Probability Table (CPT):
- Attached to each node, quantifying $P(\text{Node} \mid \text{its Parents})$.
- Acyclic Structure:
- The graph must have no cycles — you cannot follow directed edges and return to the starting node.

10.4 Example Bayesian Network — Classic 'Alarm' Style Problem



Burglary and Earthquake are independent CAUSES that influence Alarm.
Alarm ringing independently influences whether John and Mary each call.
Each node has its own CPT, e.g., $P(\text{Alarm} \mid \text{Burglary}, \text{Earthquake})$.

10.5 How Inference Works in a Bayesian Network

DEFINITION

Inference (in a Bayesian Network): The process of computing the probability of one or more unknown variables given evidence about other, observed variables in the network, using the joint probability distribution encoded by the network's structure and CPTs.

The joint probability of all variables in the network can be factored using the chain rule and the network's conditional independence structure:

$$P(X_1, X_2, \dots, X_n) = \prod P(X_i \mid \text{Parents}(X_i))$$

$$\text{Example: } P(B, E, A, J, M) = P(B) \times P(E) \times P(A|B,E) \times P(J|A) \times P(M|A)$$

10.6 Methodology — Performing Inference (Step-by-Step)

1. Define the network structure (DAG) based on domain knowledge — which variables directly influence which.
2. Specify the Conditional Probability Table (CPT) for each node given its parents.
3. Given observed 'evidence' (known values for some variables), identify the query variable(s) whose probability you want to compute.
4. Use the chain-rule factorization of the joint distribution to express the required probability.
5. Apply an inference algorithm: exact methods (Variable Elimination, Belief Propagation/Junction Tree) for smaller networks, or approximate methods (Markov Chain Monte Carlo sampling, Gibbs sampling) for larger, more complex networks.
6. Report the resulting posterior probability for the query variable(s), updated in light of the evidence.

10.7 Types of Reasoning Supported

- Diagnostic (evidential) reasoning:
- Reasoning from effects back to causes — e.g., 'JohnCalls = true; what is P(Burglary)?'
- Predictive (causal) reasoning:
- Reasoning from causes forward to effects — e.g., 'Burglary = true; what is P(JohnCalls)?'
- Intercausal reasoning:
- Reasoning between two causes of a common effect — e.g., 'Alarm = true and Earthquake = true; what is P(Burglary)?' (this can lower the probability of Burglary — a phenomenon called 'explaining away').

10.8 Advantages of Bayesian Networks

- Compactly represents complex joint probability distributions using far fewer parameters than a full joint table.
- Explicitly visualizes dependency structure, aiding interpretability and domain expert validation.
- Naturally handles missing data and uncertain evidence through probabilistic inference.
- Supports all three types of reasoning (diagnostic, predictive, intercausal) within one unified framework.

10.9 Limitations

- Constructing the correct network structure requires significant domain expertise.
- Exact inference is NP-hard for large, densely connected networks, requiring approximate methods.
- Conditional Probability Tables can grow exponentially with the number of parent nodes.

10.10 Applications

- Medical diagnosis systems (reasoning from symptoms to diseases and back).

- Spam filtering and fraud detection.
- Risk analysis in engineering and reliability systems.
- Speech recognition and natural language processing (as a precursor to Hidden Markov Models).

MEMORY TIP

Bayesian Network memory tip: 'Nodes are Nouns, Edges are Effects' — nodes are the variables (nouns), and directed edges show which variable causally affects which.

EXAM TIP

If asked to explain a Bayesian Network, ALWAYS draw a small DAG diagram (even 3-4 nodes like the Alarm example) — RGPV examiners consistently reward the visual structure heavily in graphical-model questions.

11. Exam Focus & Quick Revision

11.1 Most Important Topics (Ranked by PYQ Frequency)

1. Chi-Square Test — asked in both 2023 and 2024, numerical + conceptual variants.
2. Maximum Likelihood Test — asked in both years.
3. Regression Modelling / Analysis (variable types + simple/multiple regression) — asked in both years.
4. Multivariate Analysis — asked in both years.
5. Bayesian Modelling — asked in 2024; strong candidate to repeat.
6. t-Test (numerical application) — asked in 2024, short-note option in 2023.
7. Statistical Hypothesis Testing (general terminology) — tested indirectly across multiple questions every year.
8. ANOVA — not yet directly asked in the two available papers, but a core syllabus topic; prepare the summary table and F-statistic concept.
9. Correlation Analysis — not yet directly asked, but prepare Pearson's r formula and interpretation table.
10. Bayesian Network & Inference — newer/graphical-model topic; prepare the DAG diagram and inference concept as insurance.

11.2 Expected Long-Answer Questions (7 / 14 Marks)

- Explain statistical hypothesis testing with its key terminology and general procedure.
- What is Chi-Square test? Explain with a numerical example (goodness-of-fit or independence).
- Explain the t-test and solve a numerical to test whether a sample mean differs from a population mean.
- Explain Analysis of Variance (ANOVA) with its summary table and F-statistic formula.
- What is correlation analysis? Explain Pearson's correlation coefficient with an example.
- Discuss Maximum Likelihood Estimation in detail with a worked example (Normal or Bernoulli distribution).
- Explain the different types of variables used in Regression Modelling.
- Explain simple and multiple regression with example and assumptions.
- What do you understand by Multivariate Exploration of Data? Explain.
- What is Bayesian modelling? How does it work? Describe its advantages and disadvantages.
- Explain Bayesian Networks with a suitable example diagram, and describe how inference is performed.

11.3 Expected Short-Answer / Definition Questions

- Define Type I and Type II errors.
- State the formula for the Chi-Square test statistic.
- Differentiate a t-test from a Z-test.
- Write the one-way ANOVA hypothesis statement (H_0 and H_1).
- Define Pearson's correlation coefficient and its range.
- State Bayes' Theorem.
- Define a Bayesian Network and name its two main components.
- What is the difference between correlation and regression?

- Define Principal Component Analysis (PCA) in one line.

11.4 Frequently Repeated Concepts to Master First

- Chi-square formula + full worked contingency table example.
- MLE five-step derivation structure (distribution → likelihood → log-likelihood → derivative → estimator).
- Regression variable types table (dependent, independent, confounding, dummy) + OLS formula.
- Multivariate techniques list (PCA, cluster analysis, MANOVA, multiple regression) with one line each.
- Bayes' Theorem formula and the Prior/Likelihood/Posterior vocabulary.
- t-test worked numerical following the exact 6-step structure shown in Section 3.5.

11.5 Quick Revision Sheet

Concept	One-Line Definition / Formula
Null Hypothesis (H ₀)	Default 'no effect / no difference' claim being tested
Type I Error	Rejecting a true H ₀ (false positive), probability = α
Type II Error	Failing to reject a false H ₀ (false negative), probability = β
Chi-Square	$\chi^2 = \sum (O-E)^2/E$ — tests association/fit for categorical data
t-Test	$t = (\bar{x} - \mu) / (s/\sqrt{n})$ — compares means with small samples/unknown σ
ANOVA	$F = MSB/MSW$ — compares means across 3+ groups
Pearson's r	$\Sigma(x-\bar{x})(y-\bar{y}) / [\sqrt{\Sigma(x-\bar{x})^2} \times \sqrt{\Sigma(y-\bar{y})^2}]$ — linear relationship strength
MLE	$\hat{\theta}$ that maximizes $L(\theta) = P(\text{Data} \theta)$; solved via $d(\log L)/d\theta = 0$
Simple Regression	$y = a + bx + \epsilon$ — one predictor, straight-line model
Multiple Regression	$y = a + b_1x_1 + b_2x_2 + \dots + \epsilon$ — several predictors
R ²	Proportion of variance in y explained by the regression model
Multivariate Analysis	Techniques analyzing 3+ variables together (PCA, MANOVA, clustering, etc.)
PCA	Reduces correlated variables into fewer uncorrelated principal components
Bayes' Theorem	$P(\theta \text{Data}) = P(\text{Data} \theta)P(\theta) / P(\text{Data})$ — Posterior $\propto$ Likelihood $\times$ Prior
Bayesian Network	DAG of variables + CPTs, used for probabilistic inference among many variables

End of Unit II Study Guide. Cross-check each topic above against the latest official RGPV syllabus copy before your exam, as syllabi are occasionally revised.